

Bhavya Solanki

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Summary

B.Tech. ECE (AIML) student with hands-on experience across AI/ML, computer vision, signal processing, electronics, and programming. Completed an R&D internship at Barco, working on Python-based image processing, computer vision & automation testing. Proficient in Python, C, and MATLAB, with experience in developing AI-driven projects and applying programming and analytical skills to real-world engineering problems. Seeking opportunities to contribute to technology-driven projects across AI/ML, electronics, embedded systems, VLSI, and SDE.

Education

B.Tech in Electronics and Communication Engineering (AIML)

2023 – 2027

Galgotias University, Greater Noida

CGPA: 8.91/10

Senior Secondary Education

2023

DAV Public School, Ghaziabad

Percentage: 71.3%

Secondary Education

2021

DAV Public School, Ghaziabad

Percentage: 96.6%

Work Experience

R and D Intern | Barco Electronic Systems Pvt. Ltd.

July 2026 – Aug 2026

- Developed Python-based test automation workflows using Robot Framework, Pytest, Selenium, Playwright, and API testing for systematic software validation.
- Worked on camera integration, image processing, camera capture, OCR, display detection, image-quality analysis using OpenCV & NumPy.
- Performed debugging, test case development, defect identification, and technical documentation to support system validation and reliability.

Projects

AI-Powered Smart Energy Meter | *Python, Machine Learning, LSTM, Data Analysis*

Nov 2025 – Dec 2025

[BhavyaSolanki31/AI-Powered-Smart-Energy-Meter](https://github.com/BhavyaSolanki31/AI-Powered-Smart-Energy-Meter)

- Developed an AI-based monitoring and forecasting system using Python and LSTM models to predict energy consumption and identify usage patterns.
- Implemented anomaly detection, real-time visualization, structured testing, and data analysis to identify abnormal patterns and improve monitoring efficiency with ~95%.

DSP-Based Secure Image Authentication System | *MATLAB, DSP, Image Processing*

Sep 2025 – Dec 2025

[BhavyaSolanki31/DSP-Based-Image-Security-and-PIN-Authentication-System](https://github.com/BhavyaSolanki31/DSP-Based-Image-Security-and-PIN-Authentication-System)

- Developed a secure image authentication system using DSP algorithms, bit-plane slicing, and pseudo-random embedding, achieving PSNR >40 dB across 20+ test images while ensuring high image fidelity.
- Implemented a PIN-based authentication mechanism with <1-second validation time and ~90% authentication success rate, strengthening secure data processing and real-time system implementation.

ATS Resume Analyzer | *HTML, CSS, JavaScript*

May 2026 – June 2026

[BhavyaSolanki31/ATS-Resumer-Analyzer-Pro](https://github.com/BhavyaSolanki31/ATS-Resumer-Analyzer-Pro)

- Developed an automated resume evaluation platform that analyzes resumes against ATS standards.
- Implemented keyword matching, scoring, and improvement recommendations to support data-driven candidate assessment.
- Designed an intuitive interface for efficient document processing and report generation.

Skills

Technical Skills:

- Programming:** Python, C, MATLAB, Data Structures & Algorithms, Object-Oriented Programming
- AI/ML & Data:** Machine Learning, Deep Learning, TensorFlow, Scikit-learn, LSTM, CNN, NumPy, Pandas
- Computer Vision & Image Processing:** OpenCV, Scikit-image, Image Processing, Computer Vision
- ECE & Core:** Digital & Analog Electronics, Signal Processing, DSP, VLSI Fundamentals, Embedded Systems, Microcontrollers, IoT
- Engineering & Testing:** Testing & Validation, Technical Debugging, Defect Analysis, Root Cause Analysis, Technical Documentation
- Tools & Platforms:** MATLAB, Cadence Virtuoso, Git, GitHub, Arduino IDE, 8051, VS Code, MS Excel, MS PowerPoint

Soft Skills: Problem Solving, Communication, Teamwork, Adaptability, Attention to Detail, Analytical Thinking, Technical Documentation

Certifications

Analog ICs & Semiconductor Advancements in VLSI – NIT Delhi

Feb 2025

- Specialized training on analog ICs and semiconductor advancements in VLSI.

AI Bootcamp – NIELIT Delhi

March 2025

- Project-based sessions on supervised learning, neural networks, and real-world AI application development.